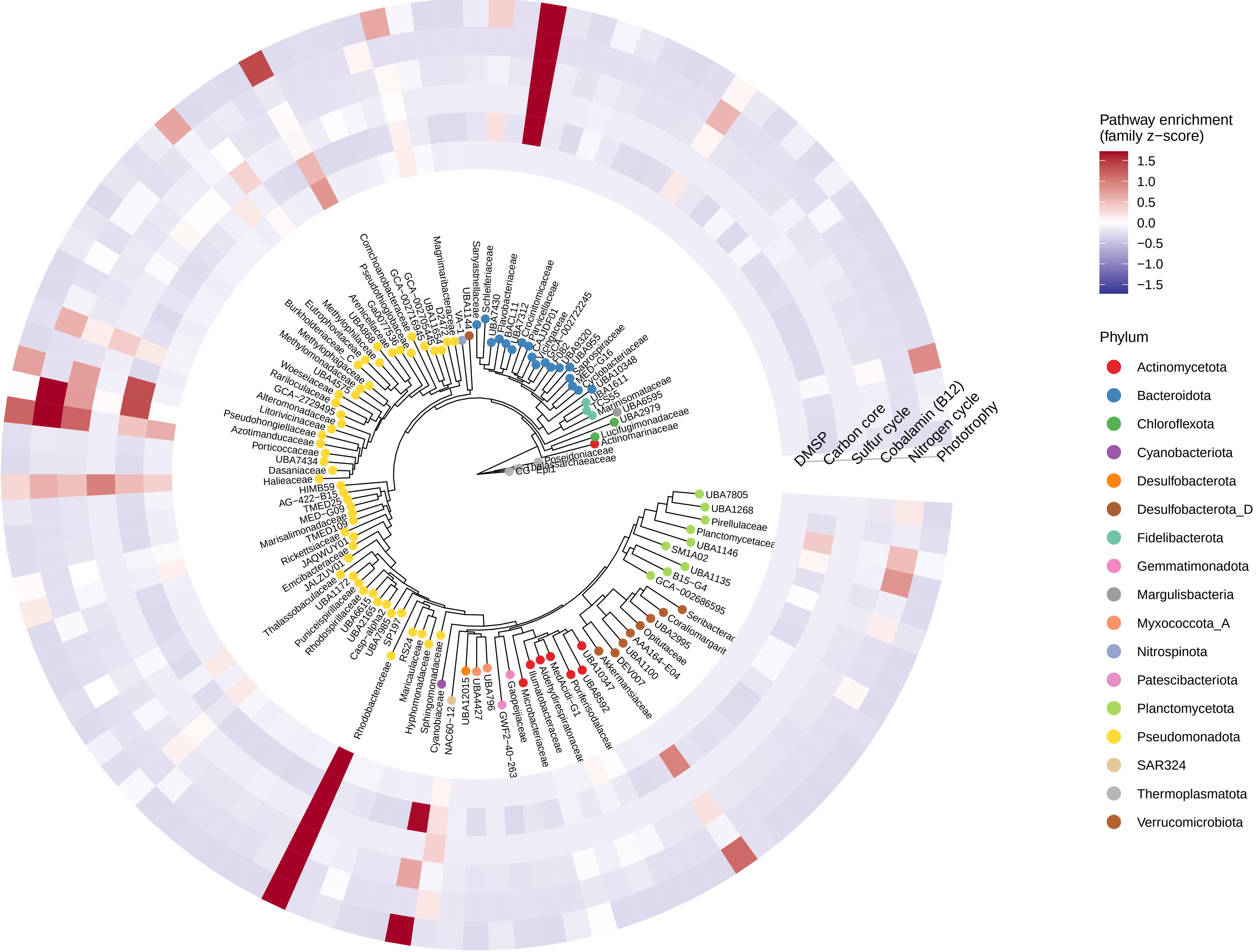
Comchoanobacteraceae
Pseudothioglobaceae
GCA−002716945
GCA−002705445
UBA11654
D2472
Magnimaribacteraceae
VA−1
UBA1144
Sanyastnellaceae
Schleiferiaceae
UBA7430
Flavobacteriaceae
BACL11
UBA7312
Crocinitomicaceae
Parvicellaceae
CAJJDF01
Vicingaceae
GCA−002722245
F082
UBA9320
UBA955
Saprospiraceae
MED−G16
Cyclobacteriaceae
UBA10348
UBA1611
TCS55
Marinisomataceae
UBA6595
UBA2979
Lucifugimonadaceae
Actinomarinaceae
Poseidoniaceae
Thalassarchaeaceae
CG−Epi1
UBA7805
UBA1268
Pirellulaceae
Planctomycetacea
UBA1146
SM1A02
UBA1135
B15−G4
GCA−002686595
Seribacterac
Coraliomargarit
UBA2995
Opitutaceae
AAA164−E04
UBA1100
DEV007
Akkermansiaceae
UBA10347
UBA8592
Poriferisodalacea
MedAcidi−G1
Aldehydirespiratoracea
Ilumatobacteraceae
Microbacteriaceae
Gaopeijiaceae
GWF2−40−263
UBA796
UBA4427
UBA12015
NAC60−12
Cyanobiaceae
Sphingomonadaceae
Hyphomonadaceae
Maricaulaceae
RS24
Rhodobacteraceae
SP197
UBA7985
Casp−alpha2
UBA2165
UBA6615
Rhodospirillaceae
Puniceispirillaceae
UBA1172
Thalassobaculaceae
JALZUV01
Emcibacteraceae
JAQWUY01
Rickettsiaceae
TMED109
Marisalimonadaceae
MED−G09
TMED25
AG−422−B15
HIMB59
Halieaceae
Dasaniaceae
UBA7434
Porticoccaceae
Azotimanducaceae
Pseudohongiellaceae
Litorivicinaceae
Alteromonadaceae
GCA−2729495
Rariloculaceae
Woeseiaceae
UBA4575
Methylomonadaceae
Methylophagaceae
Burkholderiaceae_C
Eutrophovitaceae
Methylophilaceae
UBA868
Arenicellaceae
Ga0077536
DMSP
Carbon core
Sulfur cycle
Cobalamin (B12)
Nitrogen cycle
Phototrophy
Pathway enrichment
(family z−score)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Phylum
Actinomycetota
Bacteroidota
Chloroflexota
Cyanobacteriota
Desulfobacterota
Desulfobacterota_D
Fidelibacterota
Gemmatimonadota
Margulisbacteria
Myxococcota_A
Nitrospinota
Patescibacteriota
Planctomycetota
Pseudomonadota
SAR324
Thermoplasmatota
Verrucomicrobiota